5 Two dippers S_1 and S_2, oscillating in phase with equal amplitude at a frequency of 8.0 Hz, generate waves of wavelength 6.0 cm in a ripple tank as shown in Fig. 5.1.

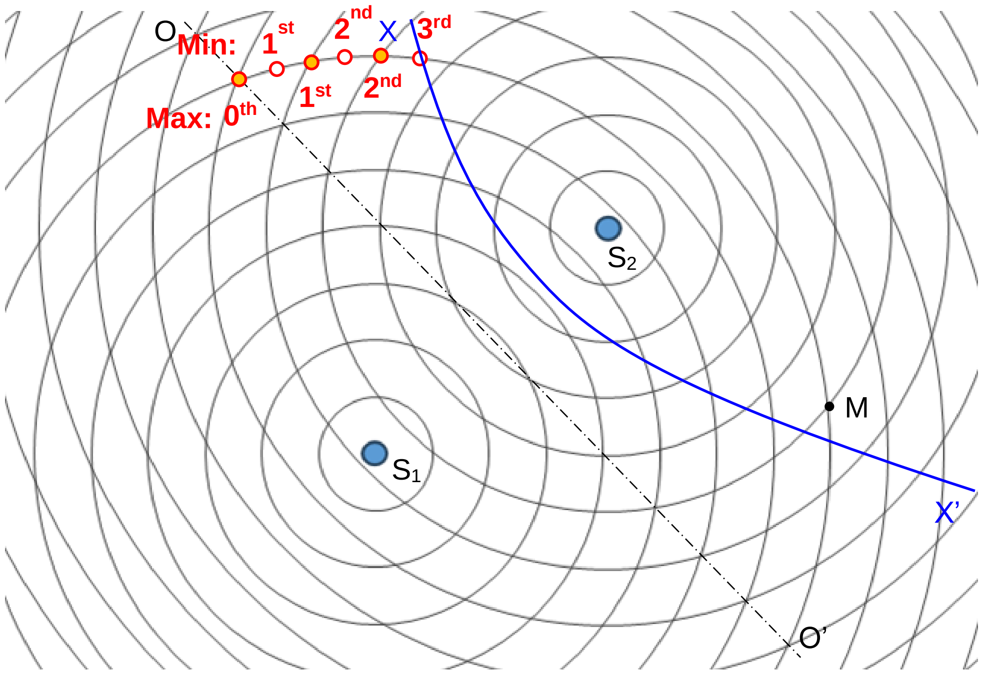

Fig. 5.1

The superposition of the waves generated produce an interference pattern of maxima and minima.

(a) State the *Principle of Superposition*.

When two or more waves meet at a point, [B1] the displacement at that point is equal to the vector sum of the displacements of the individual waves. [B1]

[2]

(b) For the waves from S_1 and S_2 meeting at point M, state

(i) their path difference,

$(8 - 5) \times 6.0 = 18.0$ cm [B1]

[1]

(ii) their phase difference.

Zero or 6π radians [B1]

[1]

(c) The waves radiate uniformly from the dippers in all directions on the surface of the water. Given that the amplitude of the wave at M when only S_1 is oscillating is 4.2 mm, deduce the amplitude of the wave at M

(i) when only S_2 is oscillating,

$$\text{Intensity} = \frac{\text{Power}}{\text{Area}} = \frac{P}{\pi s d} \propto \frac{1}{d}$$

where d is distance from dipper and s is depth of surface of water,

and Intensity $\propto$ amplitude^2

$\rightarrow$ amplitude $\propto \sqrt{\frac{1}{d}}$ [M1]

amplitude due to $S_2 = \sqrt{\frac{8}{5}} \times 4.2 = 5.3$ mm [A1]

amplitude = ... mm [2]

(ii) when both S_1 and S_2 are oscillating.

At M, the two waves are in phase $\rightarrow$ amplitude = 5.3 + 4.2 = 9.5 mm

amplitude = ... mm [1]

(d) OO' is the perpendicular bisector of S_1S_2.

(i) Draw a line on Fig 5.1 to represent the third minima from OO' and label it XX'. [1]

(ii) Explain why the amplitude of the wave along XX' is not zero.

(Due to the different distances from their respective sources,) the two waves have different amplitudes. [B1] Thus minimum amplitude = $a_1 - a_2 \neq 0$. [1]

(e) The frequency of S_1 is kept at 8.0 Hz and the frequency of S_2 is decreased slightly to 7.8 Hz.

Describe what will be observed at M.

The amplitude of the wave at M will vary from maximum to minimum and back to maximum [B1] once every 5 seconds. [B1]

....

....[2]

[Total: 11]

6 **(a)** Define *acceleration*.